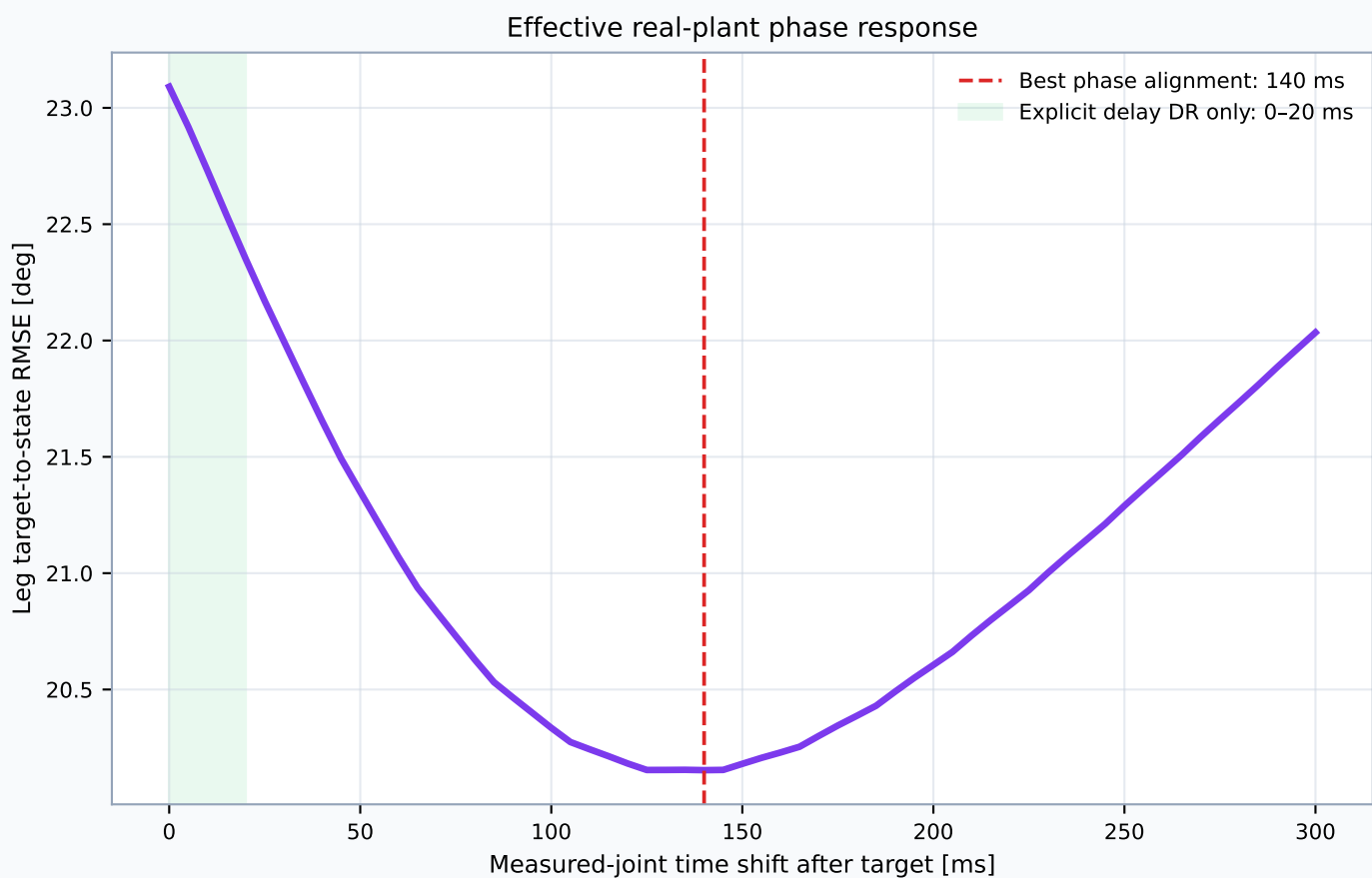
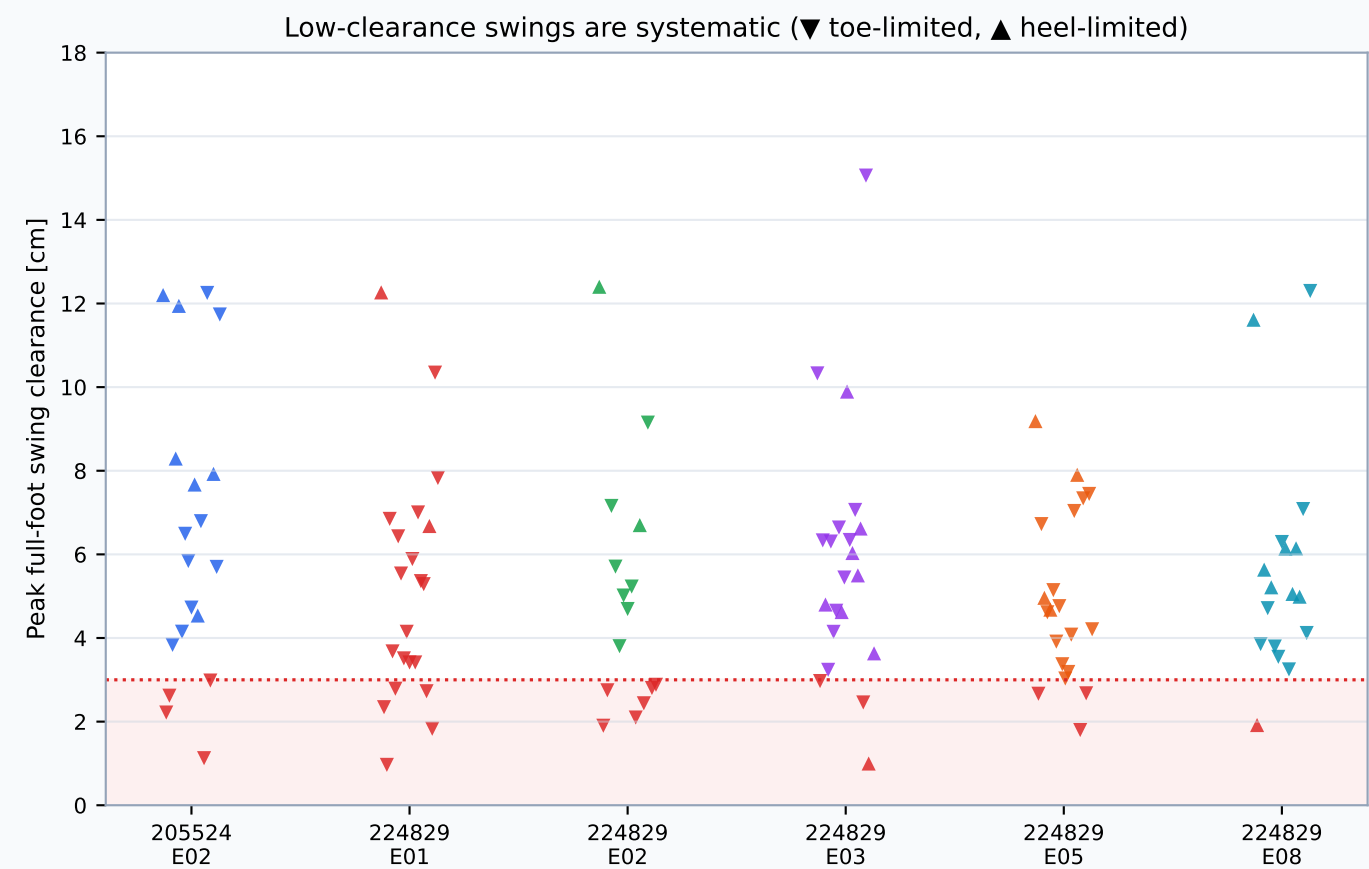
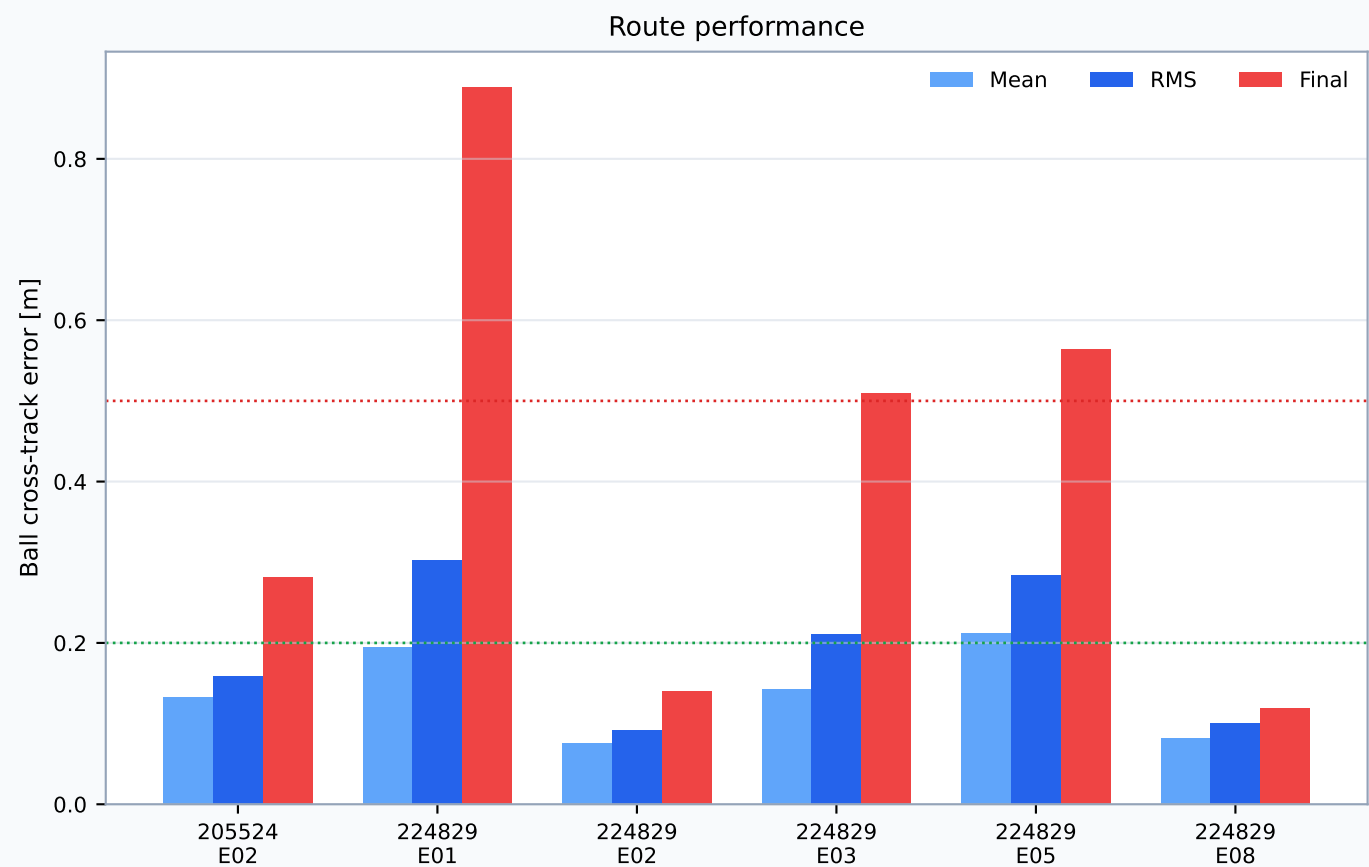
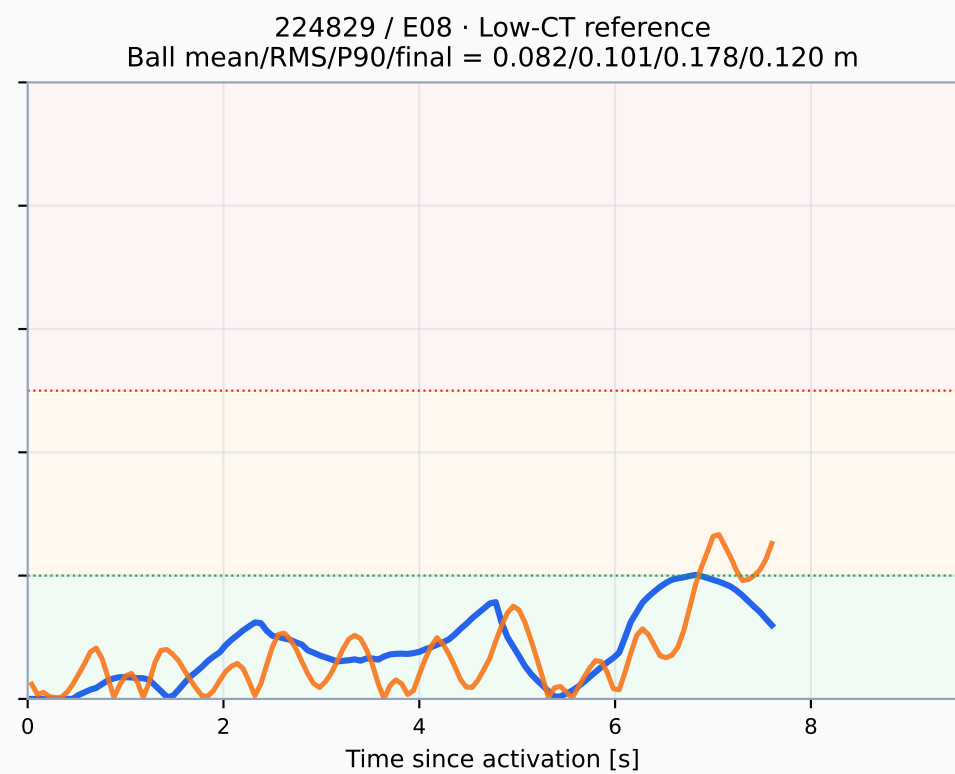
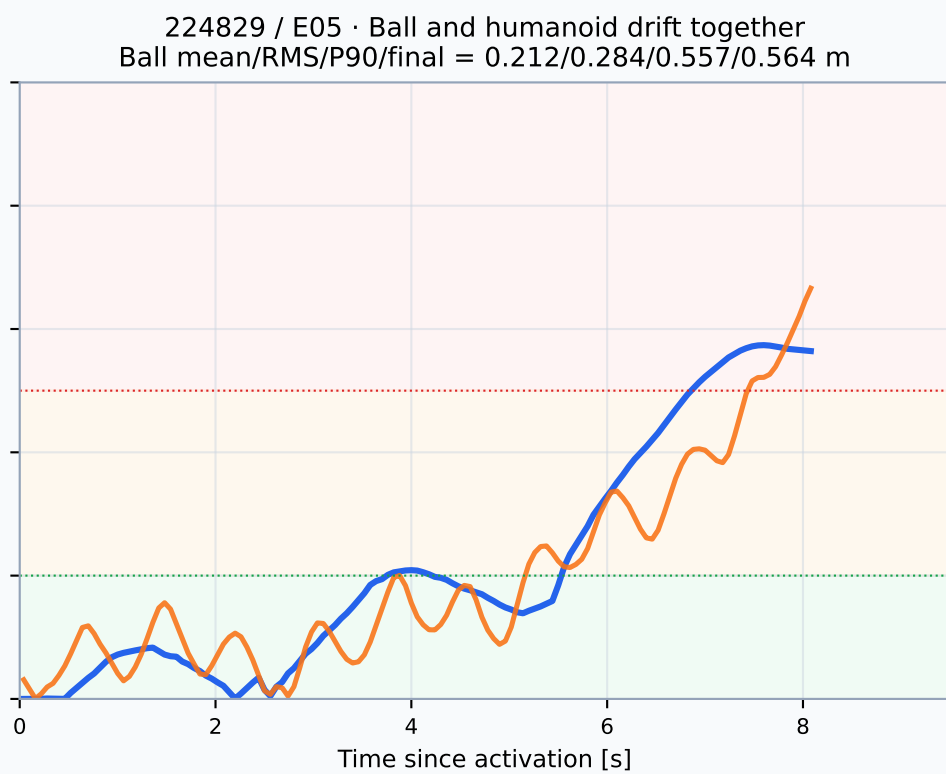
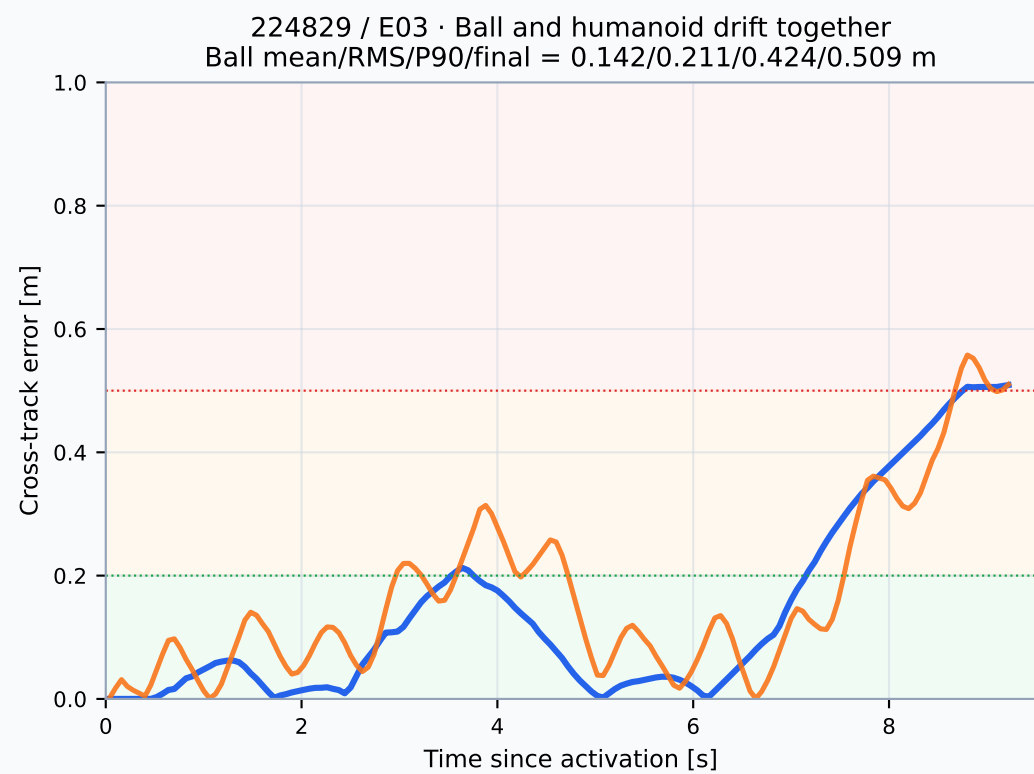
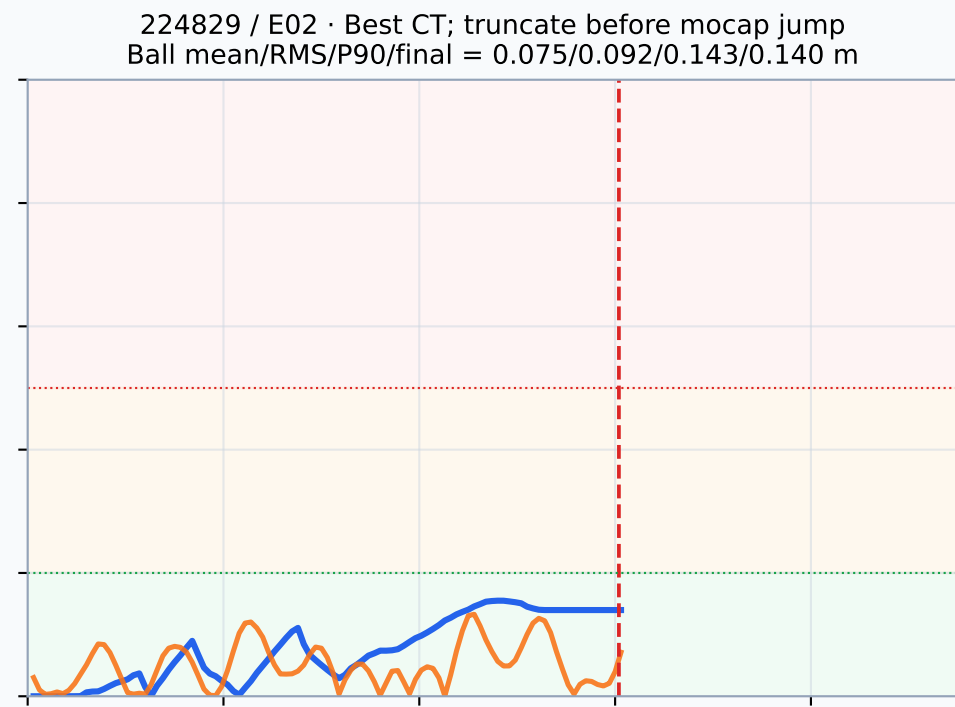
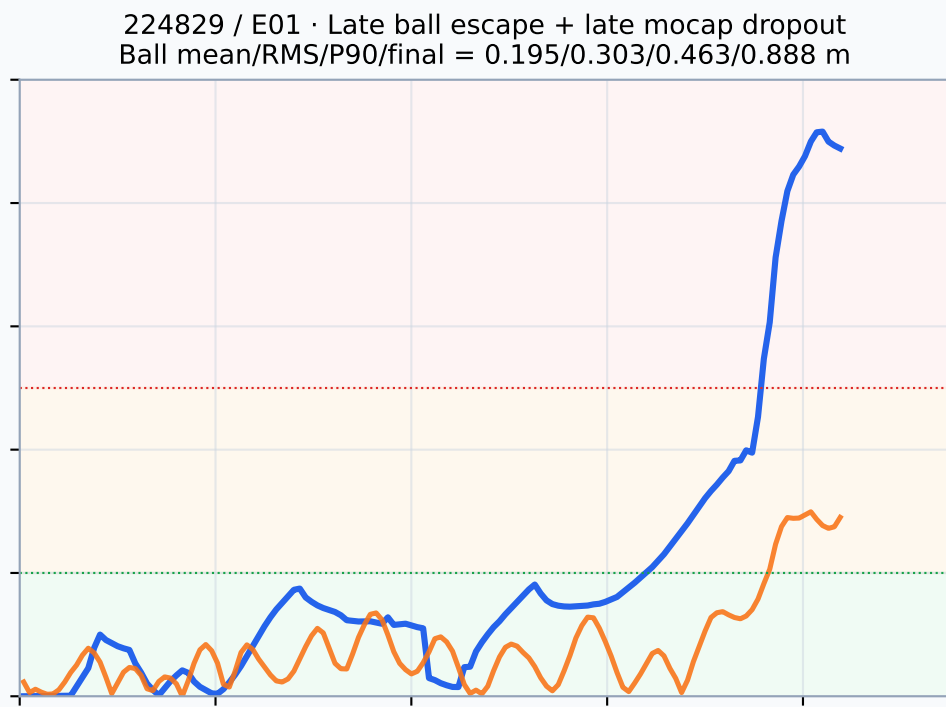
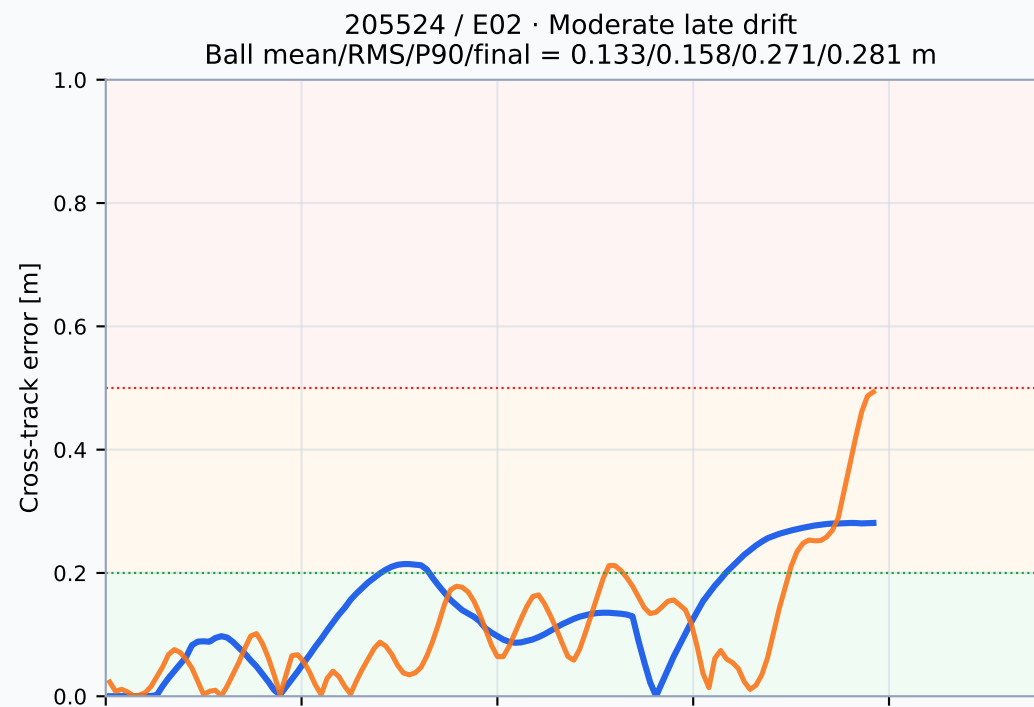


SOFTTOUCH HARDWARE FINDINGS · SIX SELECTED EPISODES



Pooled ball CT: mean 0.144 m · RMS 0.214 m · P90 0.346 m. Low-clearance swings: 22/114. Core interpretation: low toe clearance is real and systematic, while route failures split into ball escape and whole-body drift.

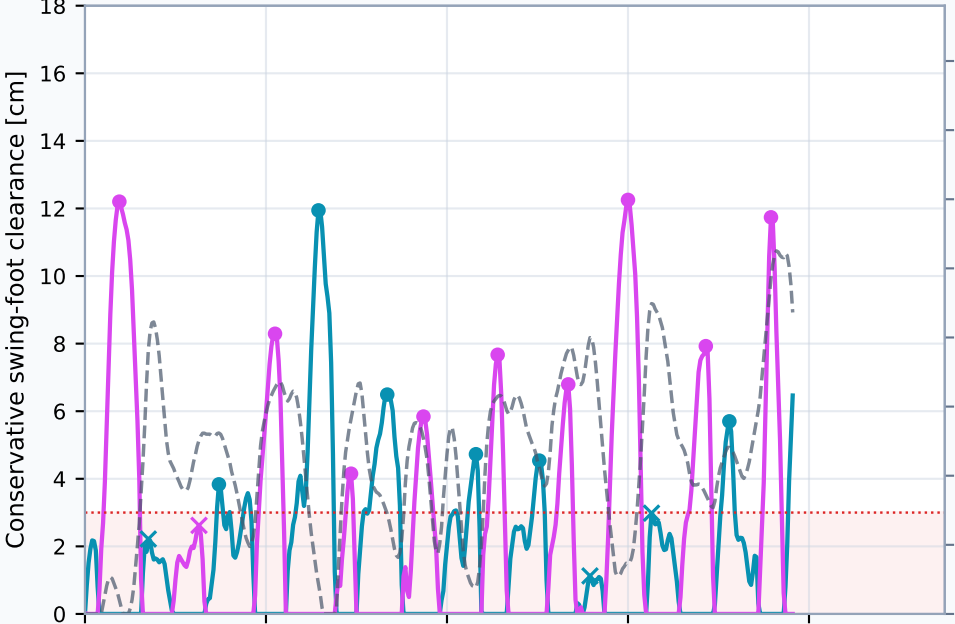
CROSS-TRACK ERROR · BALL VS. HUMANOID



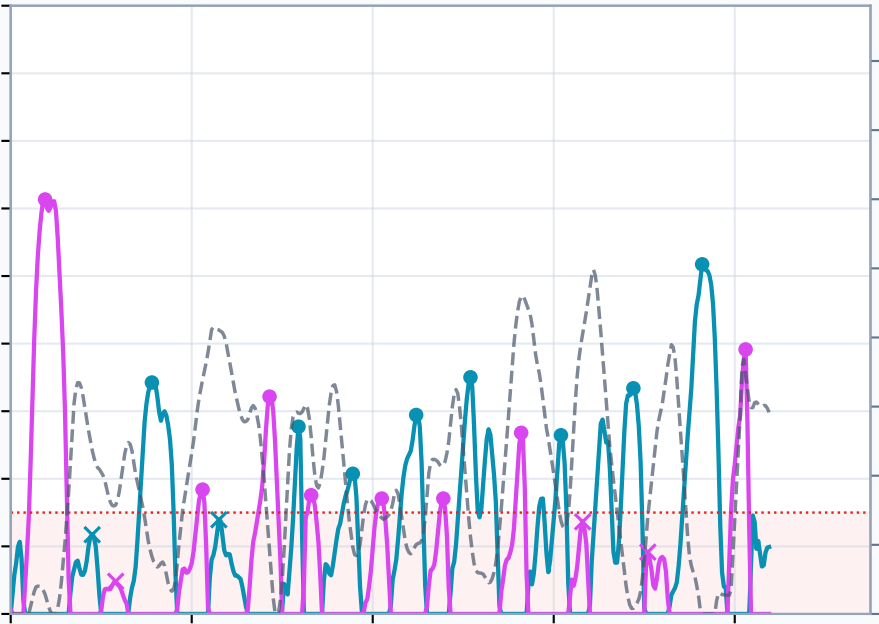
— Ball CT — Humanoid CT - - - Trusted-data cutoff

GROUND-SCRAPE EVIDENCE · FULL-FOOT MESH CLEARANCE

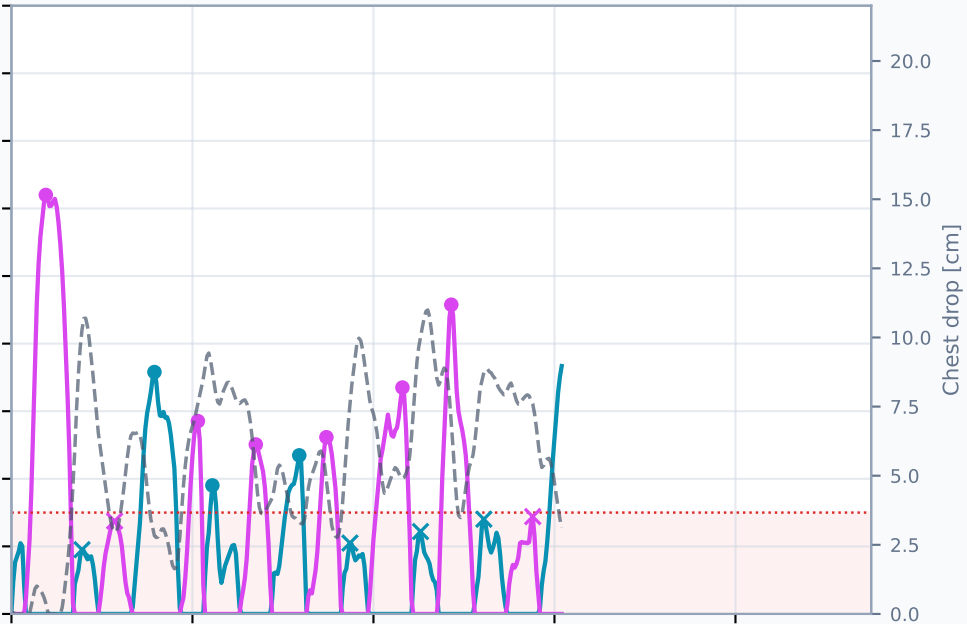
205524 / E02 · low swings 4/19 (severe <2 cm: 1)
Low swings toe-limited: 4/4 · max chest drop: 13.1 cm



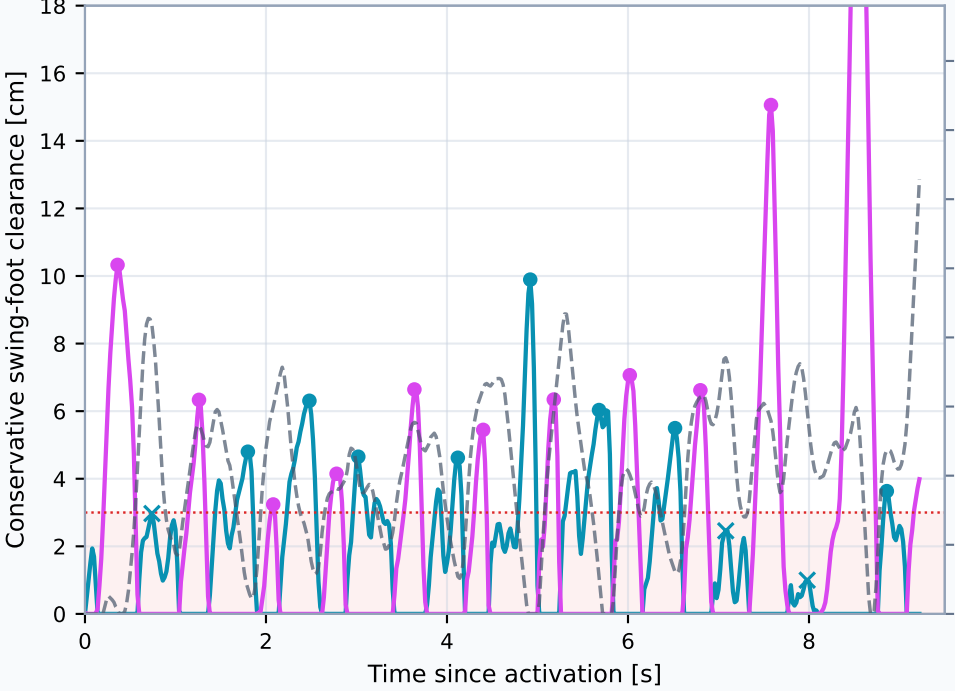
224829 / E01 · low swings 5/21 (severe <2 cm: 2)
Low swings toe-limited: 5/5 · max chest drop: 12.4 cm



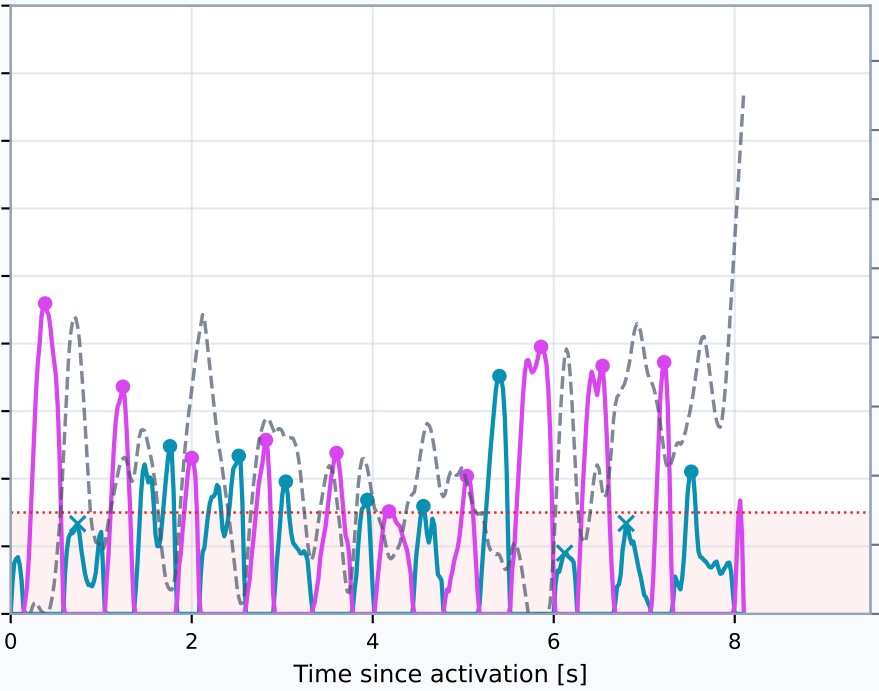
224829 / E02 · low swings 6/15 (severe <2 cm: 1)
Low swings toe-limited: 6/6 · max chest drop: 11.0 cm



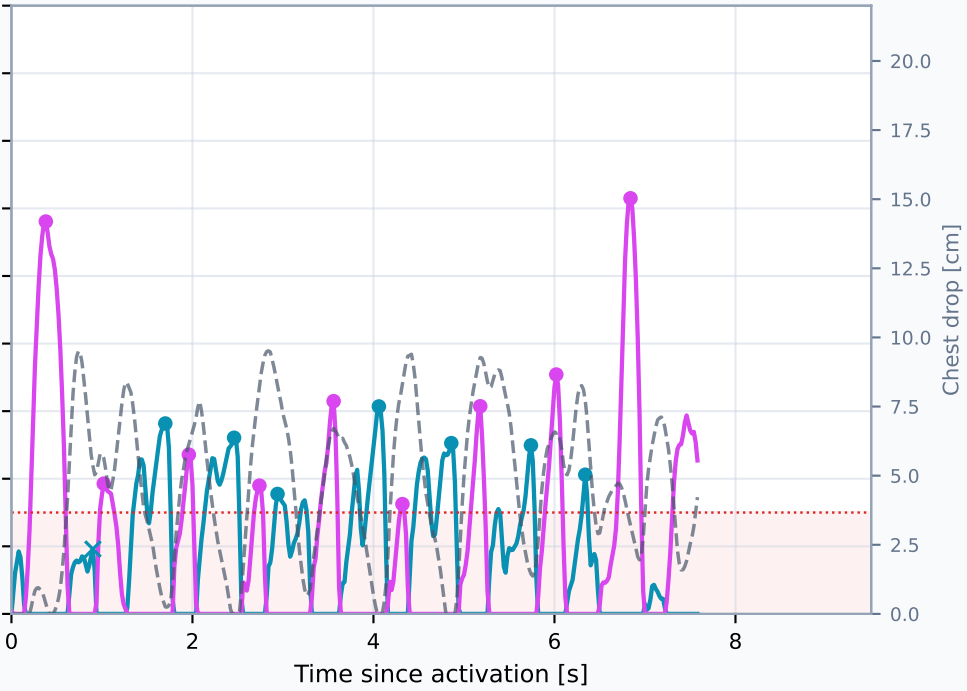
224829 / E03 · low swings 3/22 (severe <2 cm: 1)
Low swings toe-limited: 2/3 · max chest drop: 15.7 cm



224829 / E05 · low swings 3/20 (severe <2 cm: 1)
Low swings toe-limited: 3/3 · max chest drop: 18.9 cm



224829 / E08 · low swings 1/17 (severe <2 cm: 1)
Low swings toe-limited: 0/1 · max chest drop: 9.5 cm

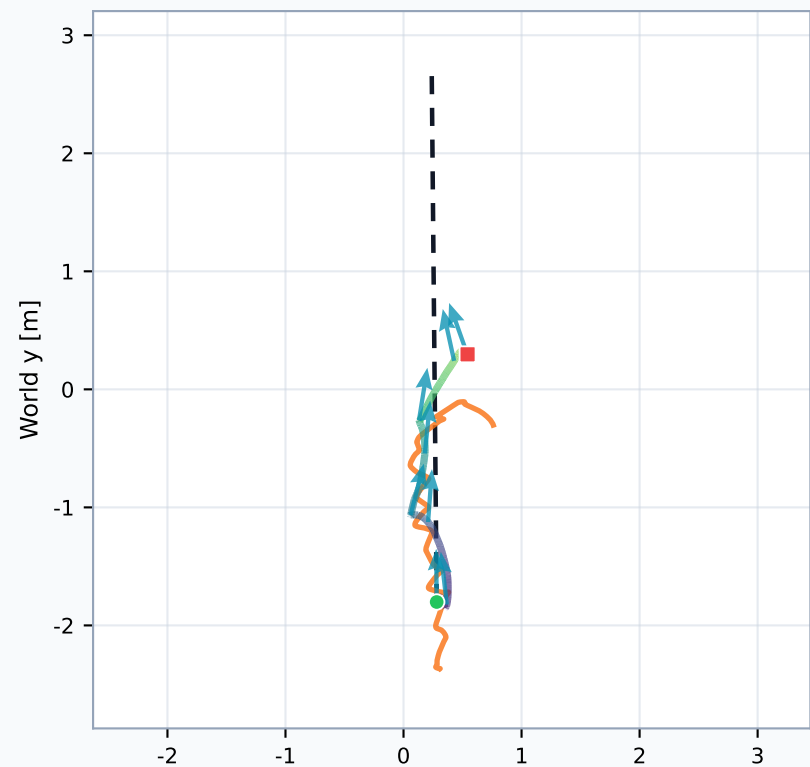


Support-plane reference = lower foot at the same instant. This cancels absolute mocap height bias. A <3 cm swing is a scrape-risk flag, not a force-sensor contact label.

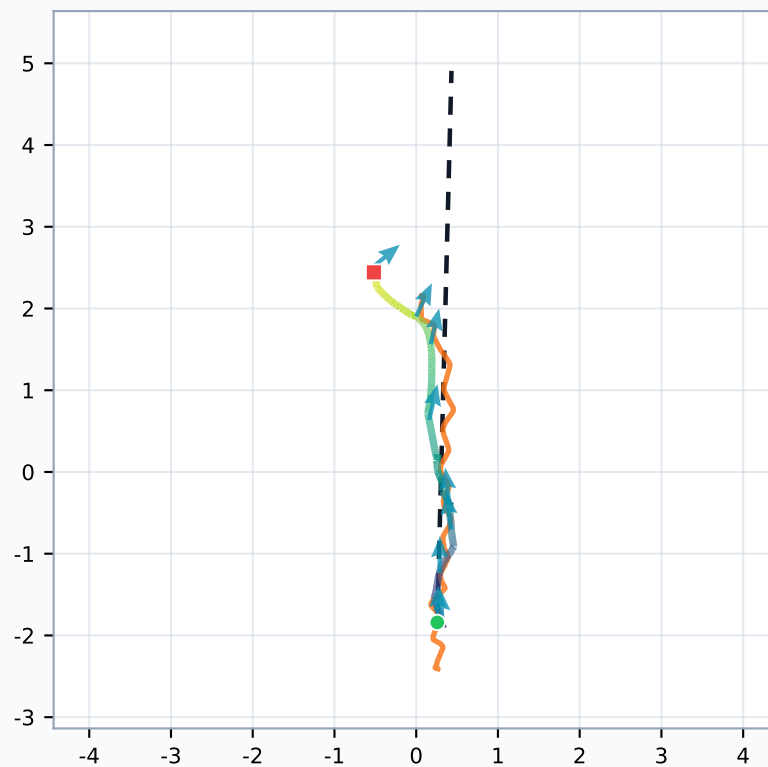
Left full-foot clearance Right full-foot clearance Chest drop from activation height Swing peak below 3 cm

FOCUSED HARDWARE EPISODES · ROUTE, BALL, AND HUMANOID

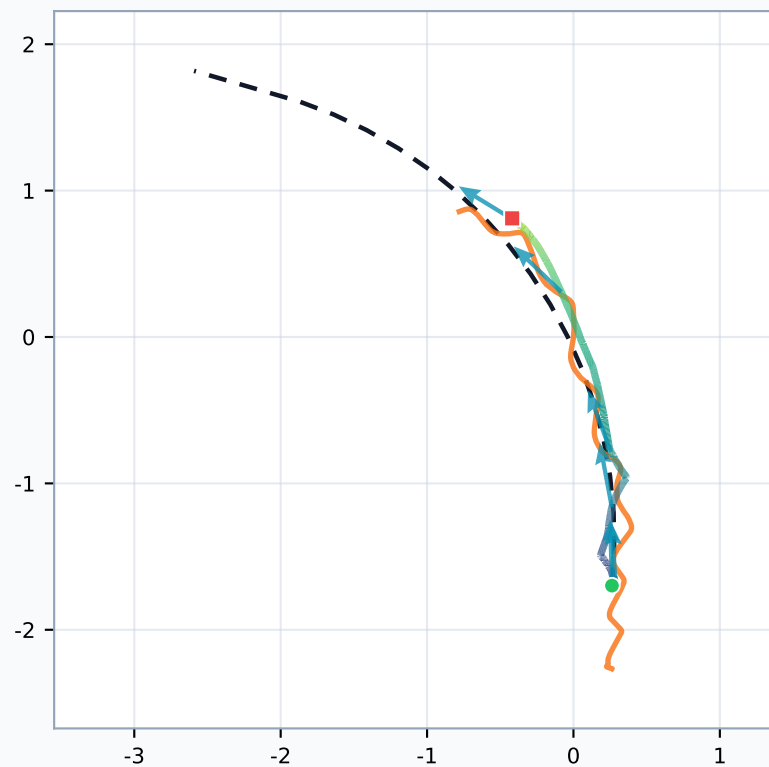
205524 / E02 · Moderate late drift
Ball CT mean/final: 0.133/0.281 m · Humanoid final: 0.494 m



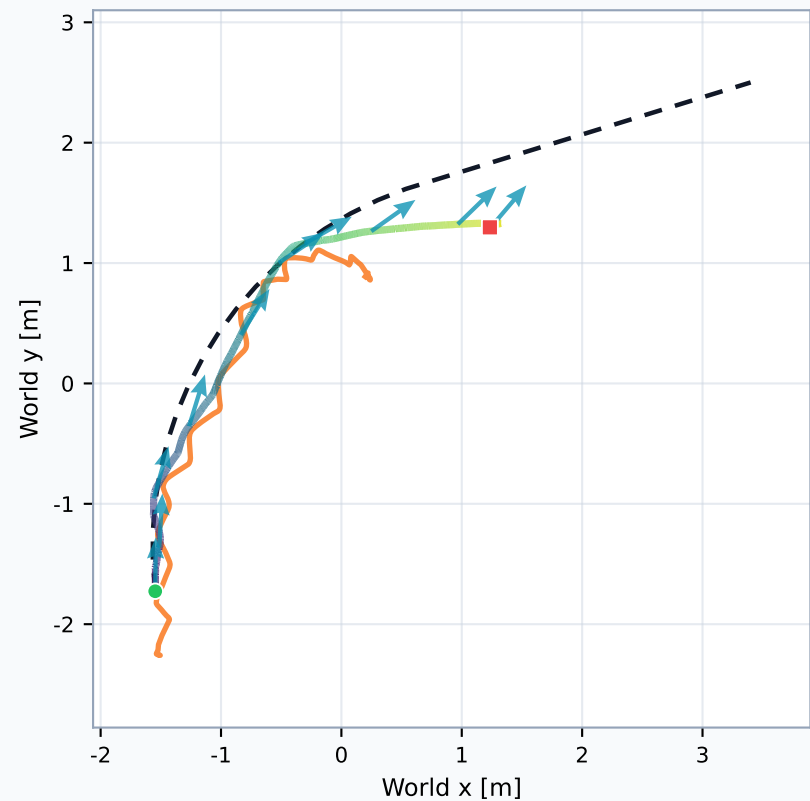
224829 / E01 · Late ball escape + late mocap dropout
Ball CT mean/final: 0.195/0.888 m · Humanoid final: 0.290 m



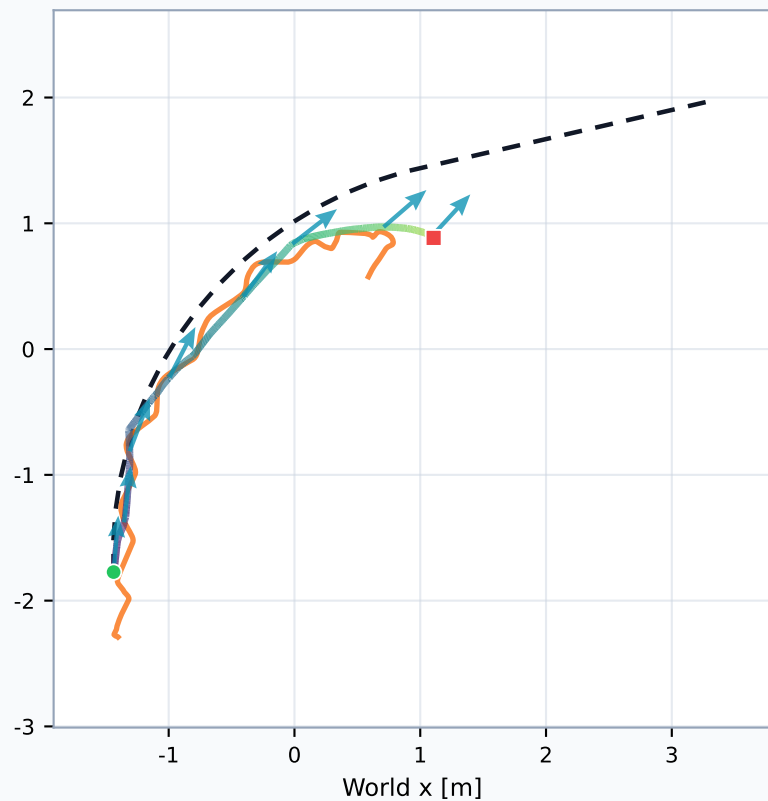
224829 / E02 · Best CT; truncate before mocap jump
Ball CT mean/final: 0.075/0.140 m · Humanoid final: 0.071 m



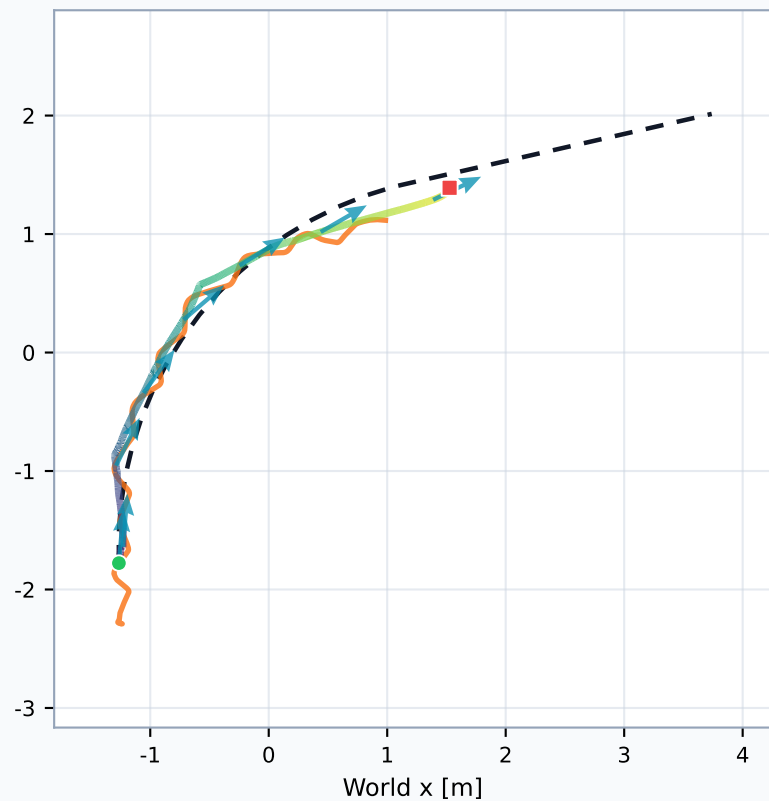
224829 / E03 · Ball and humanoid drift together
Ball CT mean/final: 0.142/0.509 m · Humanoid final: 0.510 m



224829 / E05 · Ball and humanoid drift together
Ball CT mean/final: 0.212/0.564 m · Humanoid final: 0.666 m



224829 / E08 · Low-CT reference
Ball CT mean/final: 0.082/0.120 m · Humanoid final: 0.252 m



--- Commanded route — Ball track (early → late) — Humanoid chest track — Command direction (1.0 s spacing)